

Three Types of Analytics

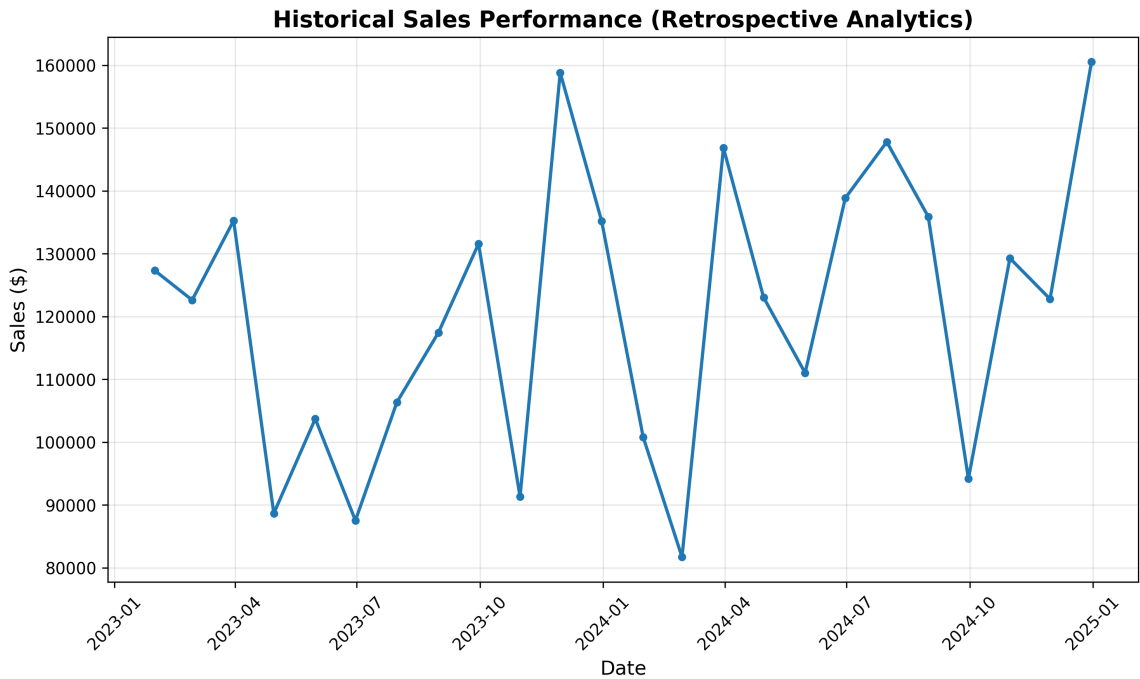
This report demonstrates the three primary types of analytics used in business intelligence and data science: Retrospective Analytics, Predictive Analytics, and Prescriptive Analytics. Each type serves a unique purpose in data-driven decision making.

1. Retrospective Analytics (Descriptive)

What is it? Retrospective analytics, also known as descriptive analytics, examines historical data to understand what happened in the past. It provides insights into trends, patterns, and performance metrics using data visualization, reporting, and statistical analysis.

Use Cases:

- Financial reporting and performance dashboards
- Sales trend analysis and revenue tracking
- Website traffic and user behavior analysis
- Customer demographics and segmentation
- Operational efficiency metrics and KPI monitoring

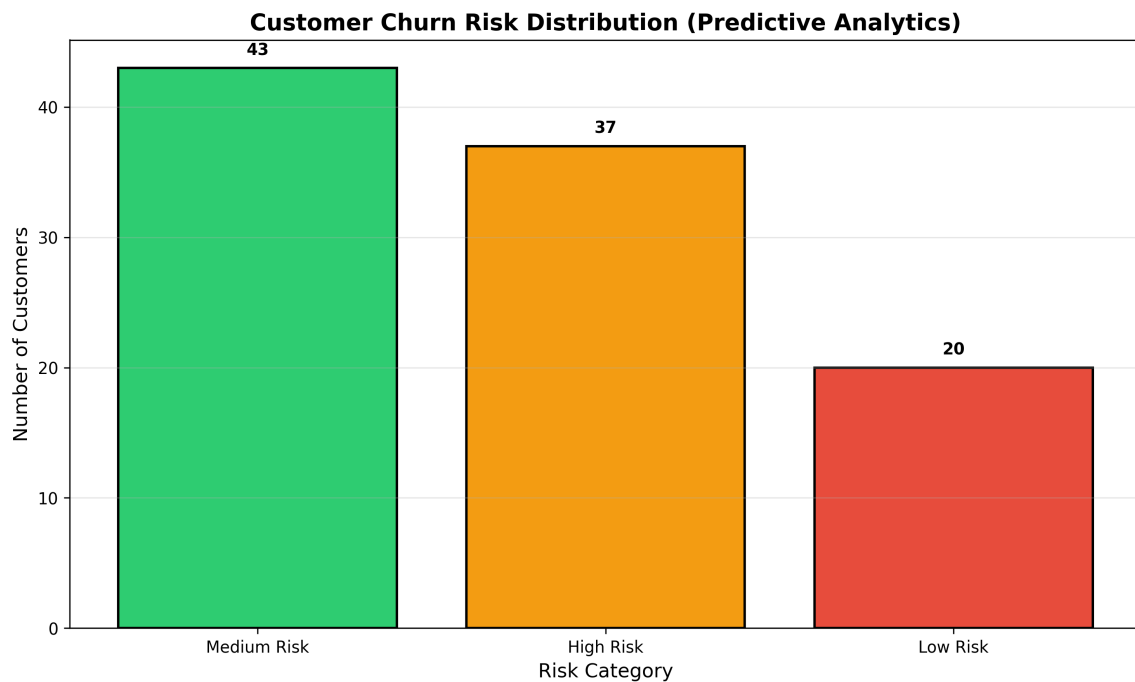


2. Predictive Analytics

What is it? Predictive analytics uses statistical models, machine learning algorithms, and historical data to forecast future outcomes. It answers the question "What is likely to happen?" by identifying patterns and relationships in data to make probabilistic predictions.

Use Cases:

- Customer churn prediction and retention strategies
- Demand forecasting for inventory management
- Credit risk assessment and loan default prediction
- Predictive maintenance for equipment failures
- Sales forecasting and revenue projections
- Healthcare outcome predictions and disease progression

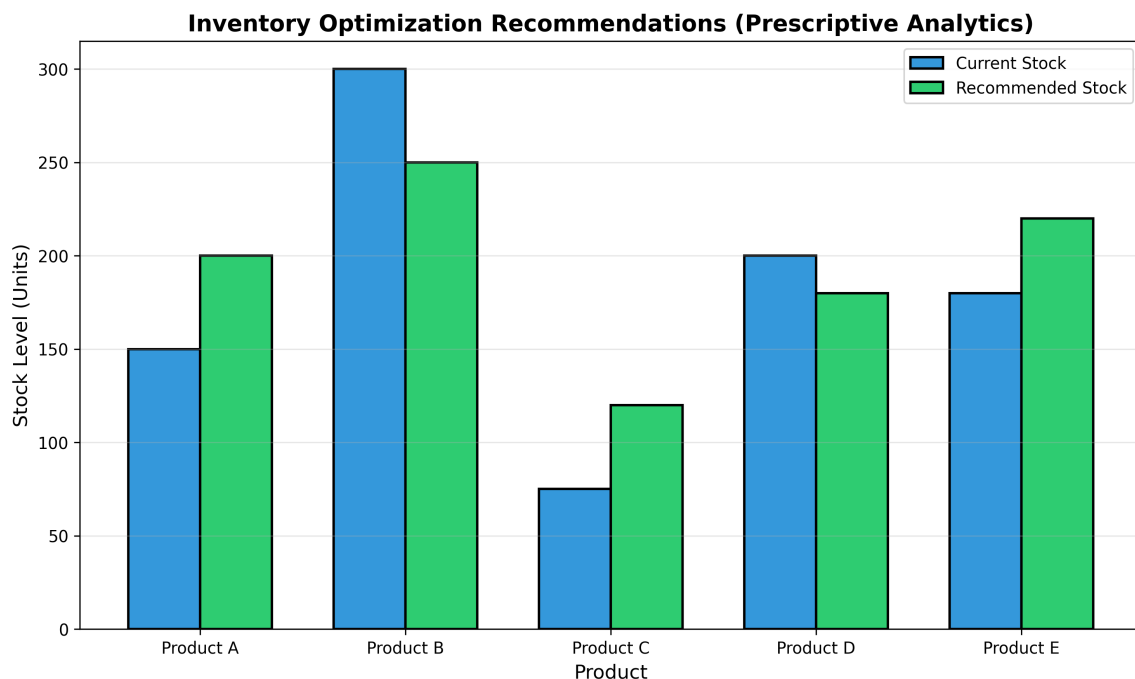


3. Prescriptive Analytics

What is it? Prescriptive analytics goes beyond prediction to recommend specific actions. It uses optimization algorithms, simulation, and decision analysis to suggest the best course of action among multiple alternatives. It answers "What should we do?" by evaluating various scenarios and their potential outcomes.

Use Cases:

- Supply chain optimization and route planning
- Dynamic pricing strategies for maximum profit
- Resource allocation and workforce scheduling
- Marketing campaign optimization and budget allocation
- Investment portfolio optimization
- Clinical decision support and treatment recommendations
- Energy grid management and load balancing



Summary

These three types of analytics build upon each other to create a comprehensive approach to data-driven decision making. Retrospective analytics tells us what happened, predictive analytics forecasts what might happen, and prescriptive analytics recommends what actions to take. Organizations that effectively leverage all three types gain significant competitive advantages through better insights and optimized decision-making processes.